

WASP-43b

R-0304

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_260321 · created 2026-07-17T21:16:03+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2461120.6913 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.197 +/- 0.015
Transit depth (Rp/Rs)^2	3.87 +/- 0.59 [%]
Orbital Inclination (inc)	79.63 +/- 0.62
Airmass coefficient 1 (a1)	0.983 +/- 0.012
Airmass coefficient 2 (a2)	0.012 +/- 0.011
Scatter in the residuals of the lightcurve fit is	1.21 %
Best Comparison Star	#1 - [356, 289]
Optimal Aperture	4.08
Optimal Annulus	10.48
Transit Duration (day)	0.0443 +/- 0.0033

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.197 ± 0.015 vs 0.1594 (deviation 2.3σ)
Duration fit vs published	0.0443 d vs 0.0512 d (deviation 1.93σ)
Mid-time O–C	2.7 min
Depth SNR	12.0
Residual scatter	1.21 %

Light curve

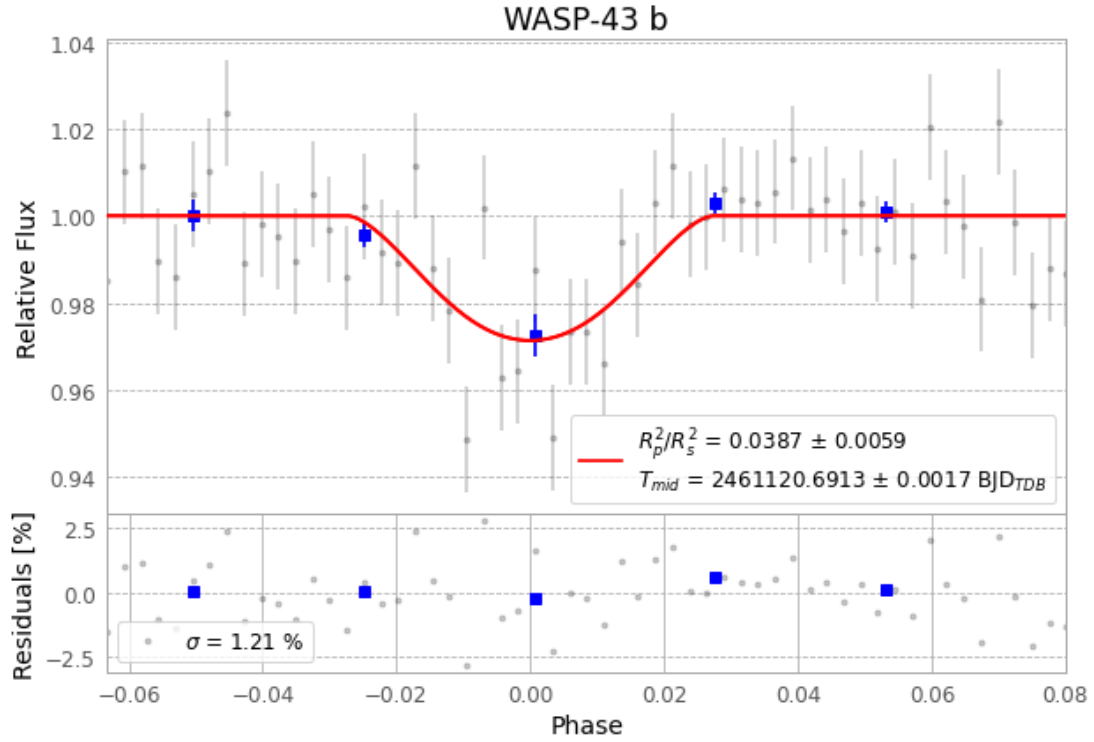


Figure 1 — FinalLightCurve_WASP-43 b_2026-03-20.png (SHA-256 4d2c074615dabd36...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [335, 161], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7fe236aa0e785e56...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 055cf2e

Data provenance

57 FITS frames (37 MB), WASP-43260321031215.FITS → WASP-43260321060015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-260321.log (4dd7452ebff7...), FinalLightCurve_WASP-43 b_2026-03-20.png (4d2c074615da...), FinalLightCurve_WASP-43 b_2026-03-20.csv (3a141bf5be6a...)

Compute resources

Wall time 120.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 21:16Z.